

Emily Terry

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EDUCATION

American University

January 2025 - May 2026

Master's, Data Science

GPA: 3.9

- TA, Advanced Machine Learning; TA, Data Management
- Coursework: Machine Learning, Applied NLP, Regression, Bayesian Statistics, Statistical Programming, Database Systems

Pennsylvania State University

August 2018 - May 2023

Bachelor's, Data Science

GPA: 3.5

- Coursework: Big Data Analytics, Cloud Computing, Intro to Artificial Intelligence, Data Structures & Algorithms, Data Mining

SKILLS & TRAININGS

- **Programming:** Python (pandas, NumPy, Matplotlib, scikit-learn, statsmodels), PyTorch, TensorFlow; SQL; R
- **Tools:** Docker, Git, Snowflake, Tableau, Power BI, R Shiny
- **NLP Specializations:** BoW/TF-IDF, Word2Vec embeddings, text classification (ML/CNN/RNN), NER, topic modeling (LDA)
- **NVIDIA Deep Learning Institute:** Accelerated Computing with CUDA Python; Accelerating End-to-End Data Science Workflows; Generative & Agentic AI; Building Deep Research Agents; and Multi-Modal Data Curation.

PROFESSIONAL EXPERIENCE

Perry International Inc.

New York, NY, USA

Data Analyst

February 2024 - January 2025

- Engineered a financial reallocation strategy by modeling cross-market KPIs (Churn rate, Seasonality Adjusted Capital Efficiency ratios) via statsmodels OLS regression, extracting predictive drivers that reduced digital footprint costs by 15%
- Architected demand-forecast models (MAPE < 17%) using XGBoost and statsmodels to optimize inventory and media-buy schedules across the supply chain, successfully reducing stock-outs by 10%
- Deployed automated ELT pipelines via custom SQL extracts, feeding optimized Tableau dashboards to eliminate data reporting latency and accelerate cross-functional strategic decisions

Koch Industries

Auburn Hills, MI, USA

Data Scientist (FP&A)

May 2023 - February 2024

- Architected modular Python ETL pipelines via pandas and NumPy to process continuous Guardian Glass production data, standardizing analytics across five North America plants and cutting data-prep overhead by 40%
- Deployed Dockerized analytics workflows connected to Snowflake, engineering real-time R Shiny dashboards that eliminated manual refresh cycles and optimized cross-departmental reporting bandwidth by 20 hours monthly
- Engineered statistical anomaly-detection models on continuous energy-usage streams utilizing Isolation Forests, extracting operational insights that drove process adjustments and reduced total utility expenses by 4%

PROJECTS

Multi-Modal ML for Molecular Property Prediction (BACE-1)

Washington, DC, USA

- Architected a reproducible multimodal pipeline predicting BACE-1 pIC50 from SMILES, enforcing strict train/test splits to cut model evaluation time by 40%
- Engineered tri-modal SMILES features leveraging Transformer NLP embeddings, CNN vision embeddings, and RDKit Sum-over-Bonds descriptors to maximize predictive density with minimal compute overhead
- Optimized PyTorch/TensorFlow deep learning models via tri-modal fusion (text + SoB + vision), improving baseline MAE/RMSE/R² by 15% through rigorous embedding-sensitivity analysis

Bank Marketing Success - Predictive Modeling & Campaign Strategy

Washington, DC, USA

- Engineered a predictive targeting framework for 64K call records, utilizing SMOTE for class imbalance (~11.3% "yes") to optimize call-center resource allocation by 25%
- Architected an automated R evaluation pipeline utilizing 10-fold cross-validation to benchmark 6 ML models, optimizing hyperparameter search space against AUC baselines
- Operationalized a Random Forest classifier (AUC 0.802), translating statistical drivers into cost-optimized campaign strategies and establishing a robust AUC/error monitoring plan

Drowsiness.TV - Real-Time Drowsiness Detection POC

State College, PA, USA

- Directed a 6-person team to architect a drowsiness-detection POC via OpenCV facial-landmark analysis, triggering automated video pauses and securing strategic incubation investment
- Engineered a custom CNN classifier via rigorous data augmentation, elevating accuracy from 85% to 94% while validating edge-case resilience across multi-user scenarios
- Deployed a React/Node/Python microservices pipeline via REST APIs, optimizing inference execution to sub-100 ms latency for real-time drowsiness detection with minimized cloud overhead